

# Yash Rawal

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## Professional Summary

Applied AI & Machine Learning Engineer serving a **1M+ user scale** by architecting enterprise agentic platforms and financial-grade ERP AI infrastructure across 120+ organizations. Specialized in core model training, embedding optimization, and low-latency agent execution frameworks. Active contributor to Spring AI, bridging machine learning models with high-throughput production systems.

## Skills

**Languages & Backend:** Java, Python, SQL, Spring Boot, FastAPI, PostgreSQL, SAP HANA, Data Engineering  
**AI & ML Engineering:** Machine Learning, Model Training & SFT, Agentic AI, MCP, Vector Embeddings  
**AI & ML Frameworks:** PyTorch, Spring AI, LangGraph, LangChain, LlamaIndex, CrewAI, AutoGen, vLLM, SAP Build, SAP Joule, GROBID, pdfplumber  
**Platforms & Models:** Amazon Bedrock, Azure OpenAI, Google Vertex AI, Claude, Gemini, Qwen, Meta Llama & Open Weights, SAP AI Core  
**Infrastructure & Security:** Docker, Kubernetes, PII Sanitization (BIO NER), CI/CD, LLM-as-Judge, Red Teaming, Cloud Foundry, Linux

## Work Experience

### TalenTeam Ltd., UK (via Bigscal Technologies)

#### AI/ML Engineer

Oct 2024 - Present

Remote

- Deployed a production-scale agentic platform and multi-agent systems serving **1M+ active ERP users** across 120+ enterprise tenants.
- Designed a capability contract layer to execute nested profiles inside a single ReAct loop, reducing execution overhead by **3x** and token consumption by **70%**.
- Trained domain-specific **Qwen2.5-7B** models on A100 GPUs and fine-tuned E5-large-v2 embeddings using PyTorch to optimize query intent mapping.
- Developed the Prompt Lab evaluation control plane to collect reasoning trajectories and human preference annotations, establishing a scalable Human Feedback (RLHF) Pipeline.
- Implemented a hierarchical RAG pipeline with PostgreSQL/PGVector and paragraph-level signatures, achieving a **90%+ Recall@10** retrieval rate and version-aware document deduplication.
- Secured agentic workflows against OWASP LLM vulnerabilities using phantom-token context propagation, runtime tool monitoring, and trained token-classification models for PII redaction.

### Healthray (Bigscal Technologies)

#### AI/ML Engineer Intern

Apr 2024 - Sep 2024

Surat, India

- Engineered clinical summarization pipelines for 116K+ medical records with 4-bit quantized Llama 3, and designed real-time WebSocket audio streaming for Indic ASR (Whisper, Vosk, NeMo).
- Researched audio digital signal processing (STFT, Mel-spectrograms, MFCC feature extraction via librosa/torchaudio) and CNN audio architectures (VGGish, ResNet), compiling custom Indic LJSpeech-format datasets.

## Projects

### Spring AI (spring-projects/spring-ai) | Open Source

Merged PR #5711 to propagate reasoningContent from AWS Bedrock and OpenAI-compatible models, and resolved Converse API prompt-caching TTL latency consistency.

Java, Spring Boot

### Modular Agent Harness & Memory System ("Brain") | Open Source

Principal Creator & Architect of an open-source plug-and-play agentic framework designed as modular, composable building blocks for flexible enterprise workflows, implementing a formal Java-based ReAct loop (Thought-Action-Observation) with step budgets, tactical pgvector tool routing, and dynamic MCP settings caching.

Java, Spring Boot, PostgreSQL, WebSockets

### AI Agent Security Red Teaming - OpenAI | Security Researcher

Designed adversarial prompt-injection pipelines targeting guardrails to evaluate security boundaries across 2,000+ agent test cases.

Python, Kaggle

## Education

### Master of Computer Applications (Artificial Intelligence & Machine Learning) - Parul University

2022 - 2024

## Certifications

Microsoft Certified: Azure AI Fundamentals (AI-900)

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Microsoft